
Decoder block × 24 (pre-norm, residual): 12.5M each, 298.9M in all (86%)
4,096
token ids
Token embedding
49,152 × 1,024
50.3M (14%)
RMSNorm
1,024
Causal self-attention
16 heads × 64, RoPE
Q, K, V, O: 4.2M
+
RMSNorm
1,024
SwiGLU feed-forward
1,024 → 2,688 → 1,024
8.3M
+
Output layer
RMSNorm, 1,024 → 49,152
tied to embedding; softmax